

Presentation companion: the full narrative of the IFC predictive modeling project

Applied Statistics Project — Presentation Preparation Notes

Abstract

This document is not a technical report: it is a speaking aid. It walks through the entire project in the order it should be told out loud, explains every number that appears on the poster, states explicitly what was deliberately left out and why, and closes with an anticipated question-and-answer section so that no reasonable follow-up question catches the presenter unprepared. Read sections 1–7 to rehearse the talk; read section 8 the night before to rehearse the defense.

1 The one-sentence thesis

Italian municipal fragility is predictable and readable from a handful of socio-economic indicators, but a residual geographic component shows that the phenomenon is not purely individual — it is also spatial.

Everything else in the project supports this sentence. If only one sentence survives from the whole presentation, it should be this one.

2 The research question and why it matters

The Italian Fragility Composite Index (IFC) is an official municipal-level index. The project asks two questions:

1. Can the IFC be statistically explained / predicted from independent socio-economic indicators (the GRINS V3 panel)?
2. If so, what actually drives municipal fragility, and is the explanation purely socio-economic or does geography play an independent role?

This matters because if a small set of readable indicators can reconstruct most of the variance in an official fragility index, that index can be monitored cheaply between official releases, and the drivers identified become candidate levers for policy.

3 The data, in one paragraph

Target: raw IFC values for 2019 and 2021, from `IFC_Final_Analysis_Sorted.xlsx`; matches the official MFI deciles 95.6% of the time, confirming it is a faithful proxy for the index actually used by policymakers. **Predictors:** the GRINS V3 municipal panel, 549 raw columns, engineered into 326 candidate covariates (counts normalised by population plus a $\log_{1p}$ transform, negative counts normalised by population only, size variables log-transformed, rates left as is, everything winsorised at the 1st/99th percentile to limit the influence of outlier municipalities). **Sample:** 15,782 observations, 7,886 unique municipalities, two years (2019, 2021). Cross-validation is always done by splitting whole municipalities into folds, never by row — splitting by row would leak the 2019 observation of a municipality into the training fold while its 2021 twin sits in the test fold, silently inflating R^2 .

4 The modeling pipeline, told as a sequence of questions

Each step exists because it answers a question the previous step left open. Tell it in this order.

Step 1 — How much can be explained at all?

A LASSO regression (10-fold CV, λ_{1se}) is run on all 326 candidate covariates. It selects 191 of them. Out-of-sample $R^2 = 0.825$ (repeated 5×5 panel-aware CV, 25 folds). This is the *benchmark*: the ceiling of what a linear model can explain with this feature set. The instructor’s first reaction was correct: 191 covariates cannot be discussed one by one in any presentation.

Step 2 — Can the same result be told with fewer variables?

An iterative pipeline reduces 191 covariates to a parsimonious set of 25, explicit and reproducible:

1. **VIF pruning**: drop the covariate with the highest Variance Inflation Factor, recompute, repeat, until all remaining VIFs are below 10 (34 covariates dropped this way).
2. **Top-N ranking**: order the survivors by standardised effect size $|\beta \cdot SD|$ and keep the top 25 — this is what makes the variables comparable to each other despite living on different natural scales.
3. **Significance filter**: any covariate that turns out non-significant once the other 24 are in the model is automatically replaced by the next candidate in the ranking, repeated until all 25 are significant.

One explicit substantive decision precedes all three steps: the raw municipal surface area is excluded upfront, on direct instructor request, because once every other covariate is expressed as a per-capita density or rate, a static geographic descriptor like total surface area is not itself a fragility driver — it is a scale variable that the rest of the pipeline already controls for implicitly.

Result: 25 covariates, every one significant at $p < 10^{-6}$, maximum VIF = 6.4. Out-of-sample $R^2 = 0.784$ — a controlled loss of only 0.041 against the 191-covariate benchmark, for an 87% reduction in model size.

Step 3 — Does geography add anything the 25 covariates miss?

A linear mixed model adds a random intercept structure on top of the same 25 fixed effects:

$$IFC_i = \beta_0 + \sum_{j=1}^{25} \beta_j X_{ji} + u_{r(i)} + v_{p(i)|r(i)} + \varepsilon_i,$$

$$u_r \sim \mathcal{N}(0, \sigma_R^2), \quad v_{p|r} \sim \mathcal{N}(0, \sigma_P^2),$$

with r indexing 20 regions and p indexing 107 provinces nested inside regions. Four variants were fit and compared (region only, region/province nested, taxonomy-based grouping, and the combination of geography and taxonomy); the nested geographic model is the best of the four. Out-of-sample $R^2 = 0.818$ — within 0.007 of the 191-covariate benchmark, recovering almost all of the accuracy lost by going parsimonious. The intra-class correlation attributable to geography is approximately 36%: more than a third of the variance left over after the 25 fixed effects is explained by which region and province a municipality sits in, not by any covariate value.

Step 4 — Is that geographic signal already exhausted, or is something still missing?

Global Moran's I on the nested model's residuals, using a symmetrised 5-nearest-neighbour spatial weight matrix:

$$I = 0.234, \quad p < 0.001.$$

This is positive and highly significant: *after* controlling for 25 socio-economic covariates and region/province random intercepts, neighbouring municipalities still share unexplained fragility. Geography is not fully absorbed by administrative boundaries. A teammate's local LISA decomposition on the same residuals, same weights, independently recovers $I = 0.236$ — confirming the global number — and locates it: 598 municipalities form significant "High-High" clusters (under-predicted, surrounded by other under-predicted municipalities), 554 form "Low-Low" clusters, and 6,418 show no significant local pattern.

An explicit, honest limitation belongs here: explicit spatial autoregressive models (SAR/CAR) were attempted as the natural next step, but did not converge within a practical runtime on this dataset size and were abandoned rather than reported with unreliable numbers. This is stated openly, not hidden.

Step 5 — Is the model statistically trustworthy, not just a lucky R^2 ?

A battery of additional diagnostics, summarised briefly because they belong to the report rather than the poster: calibration by decile (Pearson/Spearman/Lin's CCC = 0.885/0.890/0.879), an empirical 95% prediction interval of width 7.79 IFC points, AIC/BIC model comparison confirming the nested model is preferred, and residual normality diagnostics (skewness 0.35, excess kurtosis 2.31, Shapiro–Wilk $W = 0.982$) showing mild but real deviation from normality, fully documented with QQ plots in the dedicated parsimonious-extension report.

Step 6 — Is there a fundamentally different way to reach the same accuracy?

Responding directly to the instructor's remark "PCR, not PCA":

- **PCR on the 326 GRINS features:** best $R^2 = 0.787$ at $K = 100$ components — below both the parsimonious model and the benchmark. Principal components are chosen to maximise variance in X , not predictive power for Y ; the directions do not coincide here.
- **PCR on the 12 raw IFC sub-indicators** (the same components used by the descriptive PCA cascade): $R^2 = 0.987$ at $K = 12$, i.e. using *all* the information. The 1.3% residual gap is informative, not noise: it reveals that the official IFC construction includes a non-linear step (almost certainly a percentile-rank transform) beyond a linear weighted aggregation of its 12 components.
- **PLS vs PCR, head-to-head**, same 326-feature matrix: PLS (which chooses components by covariance with Y , not variance of X alone) dominates PCR by about 5 percentage points of R^2 at every component count. PLS at $K = 75$ reaches $R^2 = 0.825$, matching the 191-covariate LASSO benchmark (0.826) almost exactly.

Conclusion of this step: dimensionality reduction *can* match LASSO's accuracy, but only by trading away interpretability — 75 latent components have no individual substantive meaning, while LASSO's 191 (or the parsimonious 25) are named, signed, readable variables. LASSO is retained as the project's primary deliverable on interpretability grounds, not because it is more accurate.

5 The five substantive drivers, explained

Standardised effect sizes ($\beta \cdot \text{SD}$): the change in IFC per one standard deviation increase in the predictor, holding the rest of the model fixed.

Rank	Variable	Direction	$\beta \cdot \text{SD}$
1	Middle-income taxpayer mass	less fragile	−1.03
2	Total taxpayer density	less fragile	−0.78
3	Low-income declared income mass	more fragile	+0.57
4	Pension share of income	more fragile	+0.56
5	Small construction firms density	less fragile	−0.53

Reading: fragility is a balance between *economic dynamism* (a broad, present middle class; a dense taxpayer base; a fabric of small construction enterprises) and *economic distress* (concentration of low-income declared earners; heavy dependence on pension income rather than wages). None of the top five drivers is a demographic or environmental variable — the strongest signal is unambiguously fiscal/economic.

6 What was deliberately left out of the poster, and why

- **Descriptive PCA on GRINS and k-means clustering on the 12 IFC sub-components** (a teammate’s contribution): these are exploratory/contextual analyses that confirm fragility is geographically structured *before* any model is fit (e.g. a cross-tab against the independent GRINS urbanisation taxonomy shows Inner Italy concentrated in deciles 9–10, Metropolitan Italy in deciles 1–2). They motivate the geographic random effects but are not part of the predictive pipeline, so they live in the introduction/context, not in the main results.
- **Residual normality QQ plots and Box–Cox profiling**: real diagnostics, fully reported in the parsimonious-extension report, but too technical to read at poster distance in the few seconds a passer-by spends on any one panel.
- **The full 191-covariate coefficient table and the complete VIF-iteration log**: available in the appendix of the benchmark report; listing them on the poster would not add information a viewer could absorb on the spot.

7 The take-home conclusion, in three points

1. **GRINS predicts IFC well**: $R^2 \approx 0.82$ out-of-sample with only 25 interpretable covariates plus a geographic hierarchy — within one percentage point of a 191-covariate benchmark and of a 75-component PLS model.
2. **The story is socio-economic, not demographic or environmental**: middle-class mass, taxpayer density and small-business fabric reduce fragility; low-income concentration and pension dependency increase it.
3. **Geography matters beyond what the indicators capture**: a 36% intra-class correlation plus a persistent, locally-confirmed Moran’s $I = 0.234$ after controlling for both covariates and administrative geography show that explicit spatial models (SAR/CAR) are the natural, currently unfinished, next step.

8 Anticipated questions and prepared answers

Q: Why LASSO and not Ridge or plain stepwise selection?

LASSO performs variable selection (sets coefficients exactly to zero), which Ridge does not; this is essential because the goal is an interpretable, sparse model, not just a low-error one. An Elastic Net cross-check across $\alpha \in \{0.1, 0.25, 0.5, 0.75, 1.0\}$ showed CV error varies by less than 1% across the whole range, confirming the choice of $\alpha = 1$ (pure LASSO) is not an arbitrary or fragile decision.

Q: What is the difference between a “feature” and a “covariate” in your slides?

A feature is a candidate variable created during engineering (326 of them), before any selection. A covariate is a feature that has actually entered a fitted regression model (191 after LASSO, 25 after parsimony refinement). The words track two different stages of the same pipeline, not two different objects.

Q: Why cross-validate by municipality rather than by row?

Each municipality appears twice (2019 and 2021). Splitting by row would let one year of a municipality train the model while the other year of the *same* municipality tests it, leaking information and inflating R^2 optimistically. Splitting whole municipalities into folds removes this leakage entirely.

Q: Why exclude total surface area?

Per direct instructor feedback: once every other covariate is expressed as a per-capita density or rate, surface area is a static scale descriptor, not a fragility driver in this specification — keeping it would have reintroduced exactly the kind of non-substantive, mechanical covariate the parsimonious model is designed to avoid.

Q: Why does PCR on the 12 official IFC sub-indicators reach only $R^2 = 0.987$, not 1.0, if those are literally the index’s own components?

Because PCR (and PCA) can only recover *linear* combinations of the input variables. The residual 1.3% indicates the official IFC construction includes a non-linear step — most plausibly a percentile-rank transform applied at some stage — that a linear projection cannot fully reproduce. This was confirmed after fixing a genuine data-quality issue (duplicate municipality-year rows in the official spreadsheet, which had been silently inflating the apparent error to 1.8% before deduplication).

Q: Why does PLS beat PCR so consistently?

PCR chooses components to maximise variance within X alone; PLS chooses components to maximise covariance between X and Y . When the directions of maximum variance and maximum predictive relevance do not coincide — which the jump observed at very low component counts in the PCR trade-off curve suggests is the case here — PLS has a structural advantage that grows with the dimensionality mismatch.

Q: If PLS matches LASSO’s accuracy, why is LASSO still your main deliverable?

Accuracy is not the only criterion the project cares about. A PLS component is a linear combination of arbitrarily many original variables with no individually interpretable meaning; a LASSO coefficient is attached to one named, signed variable a policymaker can act on. The project’s brief explicitly asks for an *interpretable* benchmark, so LASSO is preferred at equal accuracy.

Q: Why was the explicit spatial model (SAR/CAR) not completed?

It was attempted in earnest; the model failed to converge within a practical runtime on a dataset of this size (15,782 panel observations with a dense municipal-level spatial weight matrix). Rather than report unreliable coefficients from a non-converged fit, the attempt is documented as an honest limitation and a clearly identified next step, supported by the Moran’s I / LISA evidence that such a model is in fact warranted.

Q: What does the 36% intra-class correlation actually mean?

After the 25 fixed-effect covariates have done their work, the *remaining* variance in IFC splits roughly 36% to “which region/province a municipality is in” (a label, not a measured covariate) and 64% to everything else (idiosyncratic municipal variation, measurement noise, and the

spatially-correlated component later confirmed by Moran's I /LISA).

Q: How confident should we be in the top-5 drivers — could they be an artefact of multicollinearity?

The parsimonious selection enforces $VIF < 10$ for every retained covariate (maximum observed VIF is 6.4), specifically to rule out this concern; all five top drivers remain significant at $p < 10^{-6}$ even after the iterative replacement step removed collinear competitors.